

Objectives

- Automate manual processes.
- Improve accuracy and reduce paperwork.
- Maintain centralized database.
- Generate reports.
- Improve security.
- Provide user-friendly interface.
- Enable future scalability.
- Automate manual processes.
- Improve accuracy and reduce paperwork.
- Maintain centralized database.
- Generate reports.
- Improve security.
- Provide user-friendly interface.
- Enable future scalability.
- Automate manual processes.
- Improve accuracy and reduce paperwork.
- Maintain centralized database.
- Generate reports.
- Improve security.
- Provide user-friendly interface.
- Enable future scalability.

Scope

The system is intended for colleges, universities, coaching institutes, clubs, and organizations depending on the selected topic. It supports registration, management modules, reporting, authentication, and administrative control. The modular architecture allows future expansion with cloud deployment, mobile applications, notifications, analytics, and AI integration. The system is intended for colleges, universities, coaching institutes, clubs, and organizations depending on the selected topic. It supports registration, management modules, reporting, authentication, and administrative control. The modular architecture allows future expansion with cloud deployment, mobile applications, notifications, analytics, and AI integration. The E-Voting Authentication System is designed to provide a secure and organized platform for conducting digital voting in colleges, universities, coaching institutes, clubs, and other organizations. The system can manage user registration, authentication, voting-related management, reporting, and administrative control. It aims to reduce manual work, improve accuracy, maintain centralized records, and enhance data security. The system can also be expanded in the future with mobile applications, biometric authentication, cloud hosting, dashboards, notifications, analytics, and AI-based features.

Methodology

- Requirement Analysis: Identifies and documents the system requirements.
- Feasibility Study: Evaluates the technical, economic, and operational feasibility of the project.
- System Design: Defines the overall structure, modules, and functionality of the system.
- Database Design: Designs the database structure using MySQL to store and manage data.
- Implementation: Develops the system using PHP and MySQL.
- Testing: Performs unit, integration, and system testing to identify and fix errors.
- Deployment: Installs and makes the completed system available for users.
- Maintenance: Updates and improves the system after deployment.
- Documentation: Prepares technical and user documentation for the project.
- Reliability: Unit, integration, and system testing help ensure that the system works correctly and reliably.

Tools and Technologies

The project uses HTML5, CSS3, JavaScript, and Bootstrap for frontend development. HTML5 is used to create the structure of the web pages, while CSS3 is used for styling and layout. JavaScript is used to add interactive functionality, and Bootstrap helps in creating a responsive and user-friendly interface. PHP is used for backend development and server-side processing, while MySQL is used to store and manage application data. Visual Studio Code is used for writing and managing the source code, XAMPP provides the local development environment, and phpMyAdmin is used to manage the MySQL database. Modern web browsers are used to run and test the application and verify its functionality.

Timeline

Week	Activity
Week 1	Requirement Analysis
Week 2	Design
Week 3	Database
Weeks 4–5	Frontend
Weeks 6–7	Backend
Week 8	Integration
Week 9	Testing
Week 10	Documentation and Final Submission

Resources

Hardware Resources

- Laptop/Computer: Used as the primary device for developing, running, testing, and managing the project.
- Internet Connection: Required for accessing development resources, documentation, software updates, and other online requirements.
- Webcam: Required where applicable, particularly if the system uses or later supports webcam-based authentication or verification.

Software Resources

- Windows: Used as the operating system for the development environment.
- XAMPP: Provides the local server environment required to run PHP and MySQL during development and testing.
- PHP: Used for developing the server-side functionality and backend of the system.
- MySQL: Used to store and manage the application's data.
- Visual Studio Code: Used for writing, editing, and managing the project's source code.
- Web Browser: Used to access, run, and test the web application and verify its functionality and user interface.

Human Resources

- Student/Developer: Responsible for analyzing requirements, designing, developing, testing, and documenting the project.
- Project Guide: Provides guidance, suggestions, and supervision throughout the project development process.
- Testers: Test the system to identify errors and verify that the required functions work correctly.
- Administrator: Responsible for managing and controlling the administrative functions of the system.

Expected Outcomes

- **Automation of Operations:** The system will automate important activities and reduce the need for manual processes.
- **Improved Efficiency:** Digital processing will make system operations faster and more organized.
- **Reduction in Human Errors:** Automated data handling and system-based processing will help reduce errors caused by manual work.
- **Instant Report Generation:** The system will support quick generation of reports, making required information easier to access.
- **Enhanced Data Security:** The system will provide improved protection and management of application data through authentication and controlled access.
- **Better Record Management:** Maintaining information in a centralized system will make data easier to manage and retrieve.
- **User-Friendly Operation:** The system will provide an interface that can be used by administrators and end users.
- **Scalable System Structure:** The system will provide a foundation that can be extended with additional features in the future.
- **Mobile Application Support:** The system can be extended to support mobile applications in the future.
- **Biometric Authentication:** Biometric authentication can be considered as a future enhancement to strengthen user verification.
- **Cloud Hosting:** The system can be migrated to cloud hosting in the future for broader deployment and accessibility.
- **Dashboards:** Future dashboards can provide a more organized view of system information and activities.
- **Analytics:** Analytics can be added in the future to help analyze system data and generate useful insights.

References

1. Roger S. Pressman
- Software Engineering.
2. Ian Sommerville - Software Engineering.
3. PHP Documentation.
4. MySQL Documentation.
5. W3Schools.
6. MDN Web Docs.
7. OWASP (for security where applicable).

References

1. Roger S. Pressman - Software Engineering.
2. Ian Sommerville - Software Engineering.
3. PHP Documentation.
4. MySQL Documentation.
5. W3Schools.
6. MDN Web Docs.
7. OWASP (for security where applicable).
8. NIST Face Recognition Technology Evaluation (FRTE/FRVT):
<https://pages.nist.gov/frvt/html/frvt11.html>
9. OpenCV Documentation:
<https://docs.opencv.org/>